

# Dermatologic Manifestations of Chronic Kidney Disease

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Cutaneous disorders are common in patients with chronic kidney disease (CKD). Many of these cutaneous disorders are caused by the underlying kidney disease, whereas others relate to the severity and duration of uremia. CKD patients have a high incidence of skin lesions associated with cutaneous aging, including wrinkling, senile purpura, actinic keratoses, and hair loss.

Improved treatments in dialysis patients have resulted in changes in the frequency and types of skin disorders observed in conjunction with end-stage kidney disease (ESKD). Dermatologic conditions such as uremic frost, erythema papulatum uremicum, uremic roseola, and uremic erysipeloid now rarely occur. Pigmentary alterations, xerosis, ichthyosis, half-and-half nails, acquired perforating dermatosis, bullous dermatoses, pruritus, and calcific uremic arteriopathy (CUA) remain prevalent, whereas nephrogenic systemic fibrosis (NSF) is a rare and waning entity now that the causative agent has been identified (Fig. 91.1). The last four prevalent skin disorders are the focus of this chapter because they are associated with significant morbidity or mortality and/or represent an ongoing diagnostic or therapeutic challenge.

## UREMIC PRURITUS

### Clinical Manifestations

Uremic pruritus (UP) is a frequent symptom of ESKD, with a prevalence ranging from 22% to 48%. Although its incidence in adult dialysis patients declined as a result of improved dialysis efficacy and the introduction of so-called “biocompatible” dialysis membranes, UP still causes serious discomfort and skin damage, often associated with disturbance of day and night rhythm, sleeping disorders, depression, anxiety, and diminished quality of life.<sup>1,2</sup> The intensity and spatial distribution of UP vary significantly among patients and over time throughout the course of kidney disease. Excoriations, induced by uncontrollable scratching with or without superimposed infection, occur in severely affected patients and, rarely, lead to prurigo nodularis—a treatment-resistant lichenified or excoriated papulonodular chronic skin eruption (Fig. 91.2). The most frequently involved body areas are the back, limbs, chest, and face; 20% to 50% of patients report generalized pruritus.<sup>3</sup>

### Pathogenesis

Factors thought to contribute to UP include parathyroid hormone (PTH) and divalent ions (calcium, phosphate, magnesium) because itching is a frequent symptom accompanying severe secondary hyperparathyroidism and elevated calcium-phosphate product. However, the lack of consistent correlations between serum and skin levels of PTH, calcium, phosphorus, and magnesium with the severity of UP indicates that other factors also contribute to its development. The number of dermal mast cells is increased in uremic patients, and higher tryptase and histamine plasma concentrations are reported

in severe cases. Histamine release is triggered by substance P, a neurotransmitter involved in itch sensation. The role of elevated serotonin (5-hydroxytryptamine [5-HT<sub>3</sub>]) levels in dialysis patients with UP is debated; clinical trials using a selective inhibitor of 5-HT<sub>3</sub> have yielded conflicting results (see later discussion). Xerosis is a common skin problem in dialysis patients (60%–90%) that predisposes to UP. Skin dryness is caused by primary dermal changes associated with uremia, such as atrophy of sweat glands with impaired sweat secretion, disturbed stratum corneum hydration, sebaceous gland atrophy, and abnormal arborization of free cutaneous nerve fiber endings.

There are two major hypotheses concerning the pathogenesis of UP. The opioid hypothesis proposes that UP is caused by overexpression of opioid  $\mu$ -receptors and decreased  $\kappa$  opioid receptor expression in dermal cells and lymphocytes. Consistent with this hypothesis, activation of the  $\kappa$  opioid system using a  $\kappa$ -receptor agonist was efficient in reducing UP in a mouse model. In contrast, the immune hypothesis considers UP an inflammatory systemic disease rather than a local skin disorder. Studies examining the beneficial effects of ultraviolet B (UVB) light exposure on pruritus showed that UVB attenuates the development of Th1 helper cells in favor of Th2 helper cells. Indeed, the number of CXCR3-expressing and interferon- $\gamma$ -secreting CD4<sup>+</sup> cells (indicating Th1 differentiation) is significantly increased in the circulation of dialysis patients with UP compared with those without. Levels of serum markers of inflammation, such as (high sensitivity) C-reactive protein, interleukin (IL)-2, IL-31, and IL-6, are also higher in patients with UP.

### Treatment

Common causes of UP in CKD and dialysis patients, such as primary skin disorders (e.g., urticaria, psoriasis, atopic and contact dermatitis), liver disease (e.g., hepatitis), and endocrine diseases (e.g., hypothyroidism, diabetes mellitus) should be ruled out and adequately treated. The treatment approach to UP is shown in Fig. 91.3.

### Optimizing Dialysis and Mineral Metabolism Therapy

Optimizing dialysis biocompatibility and efficacy, using medium cut-off dialyzers, and improving nutritional status may reduce the prevalence and severity of UP. Controlling calcium and phosphorus serum concentrations by short-term use of dialysate with low calcium and magnesium concentrations has sometimes ameliorated UP symptoms but may lead to worsening of renal osteodystrophy in cases of prolonged use. Parathyroidectomy is not advocated for relief of UP because no consistent beneficial effects have been demonstrated.

### Skin Emollients

Emollients continue to be the primary treatment given by nephrologists, although the overall efficacy of this approach is uncertain. Simple emollients without perfumes or other additives are preferred. Continuous bath oil therapy containing polidocanol, a mixture of



**Fig. 91.1** Cutaneous Disorders in Patients With End-Stage Kidney Disease. (A) A spectrum of pigmentary alterations occurs in dialysis patients, with brownish hyperpigmentation in sun-exposed areas being the most prevalent. (B) Xerosis, a dry or roughened skin texture, is seen in up to 75% of dialysis patients. (C) Acquired perforating dermatosis affects approximately 10% of the dialysis population. The lesion is usually asymptomatic and consists of grouped dome-shaped papules and nodules, 1 to 10 mm in diameter. The trunk and the extremities are most commonly involved. (D) Half-and-half nails (also termed *red and white nails*) occur in as many as 40% of patients on dialysis. The nails exhibit a whitish or normal proximal portion and an abnormal brown distal portion.

monoether compounds of lauryl alcohol and macrogol, seems to help some patients. Sericin, a biopolymer protein from the silkworm (*Bombyx mori*), contains 32% serine, is the main amino acid of the natural moisture factor in human skin, and suppresses the release of proinflammatory cytokines. Sericin effectively reduced UP in a randomized, double-blind, placebo-controlled 6-week study in 50 dialysis patients.<sup>4</sup>

### Antihistaminic Drugs

Classic antihistamines have similar efficacy compared with emollients; newer, second-generation antihistamines (e.g., desloratadine) might

have some effect in UP. Ketotifen (2–4 mg/day), a putative mast cell stabilizer, was beneficial in one small study.

### Phototherapy

UV light, especially UVB (wavelength 280–315 nm), is effective for treatment of UP and is well tolerated except for occasional sunburn. The duration of the antipruritic effect of thrice-weekly total body UVB therapy (total of 8–10 sessions) is variable but may last for several months. Potential carcinogenic effects of UV radiation require serious consideration, and its prolonged use (>12 weeks) is contraindicated in patients with a fair complexion (skin phototypes I and II).





Fig. 91.2 Prurigo nodularis. (Courtesy I. Macdougall, London, UK.)

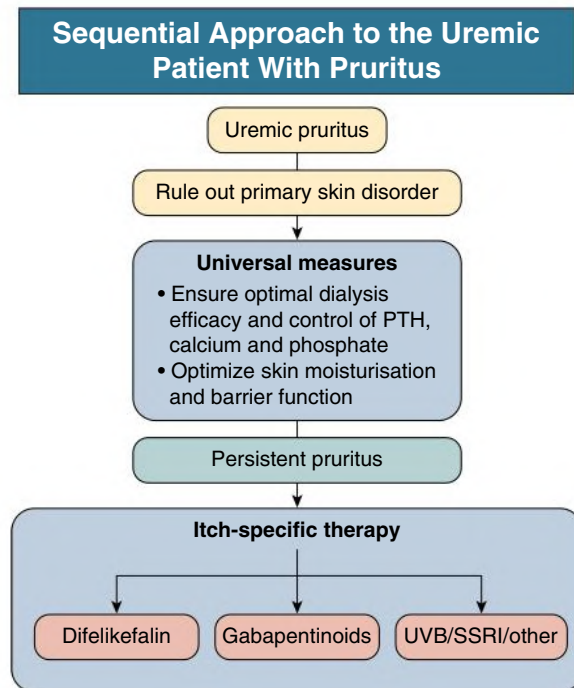


Fig. 91.3 Sequential approach to the uremic patient with pruritus. PTH, Parathyroid hormone; SSRI, Selective serotonin reuptake inhibitor; UVB, Ultraviolet B.

### 5-Hydroxytryptamine Antagonist

Ondansetron, a selective 5-HT<sub>3</sub> antagonist, was used successfully in a small study in peritoneal dialysis patients. However, a subsequent larger randomized, placebo-controlled study failed to show superiority over placebo in patients on hemodialysis (HD).

### Opioid Receptor Agonists

A  $\kappa$ -opiate receptor agonist, nalfurafine, given intravenously after HD, was tested in two randomized, double-blind, placebo-controlled trials involving 144 patients. Itching intensity, excoriations, and sleep disturbances were significantly reduced in patients receiving the active compound without an excess of drug-related side effects compared with placebo.<sup>5</sup> Oral daily doses of 2.5 to 5

mg nalfurafine were equally effective in reducing UP scores compared with placebo during a 2-week treatment period in 337 dialysis patients.<sup>6</sup> A subsequent 1-year study ( $n = 211$ ) confirmed the sustained beneficial effect of 5 mg nalfurafine on UP, with the most frequently reported adverse events being insomnia (19%), constipation (7%), and increased blood prolactin concentrations.<sup>7</sup> Difelikefalin, a peripherally restricted and selective kappa opioid receptor agonist, has been approved for treatment of moderate to severe pruritus associated with CKD in adults undergoing hemodialysis. This drug reduced UP intensity in a significantly larger proportion of dialysis patients (51.9% vs. 30.9% in controls) than placebo when administered thrice-weekly intravenously for 12 weeks.<sup>8</sup> UP-related quality of life improved in the active treatment arm, but diarrhea, dizziness, and vomiting were more commonly observed.<sup>8</sup>

### Gabapentin

Gabapentin, an anticonvulsant drug administered after dialysis (300 mg), was effective in reducing UP. A reduced dose is required if it is given chronically to patients with ESKD because it has a narrow therapeutic window and can accumulate and cause neurotoxic side effects.<sup>9</sup> The UP score was significantly reduced in two controlled studies after 4 weeks of gabapentin therapy at a dose of 100 to 300 mg administered thrice weekly after dialysis. Gabapentin was well tolerated; no patients stopped therapy because of side effects, which were limited to dizziness, somnolence, fatigue, and nausea. Similar effects were demonstrated in a smaller study using pregabalin and recent randomized trials comparing gabapentin with doxepin and ondansetron.<sup>10,11</sup>

## IMMUNOMODULATORS AND IMMUNOSUPPRESSIVE AGENTS

A 7-day course of thalidomide reduced the intensity of UP by up to 80% in a placebo-controlled crossover study of 29 HD patients. Because of its strong teratogenic properties, thalidomide should be reserved for therapy-resistant severe UP in individuals outside the reproductive-age category. Adverse effects of thalidomide, such as peripheral neuropathy and cardiovascular side effects, limit its longer use. A prospective single-center study of 25 chronic dialysis patients with UP demonstrated that 6 weeks of treatment with tacrolimus ointment (0.1%) significantly reduced the severity of UP. Tacrolimus was well tolerated in this trial and caused no detectable systemic exposure or side effects.<sup>12</sup> However, a subsequent, smaller vehicle-controlled trial showed equal relief of UP with vehicle and tacrolimus. The risks of long-term topical use of these agents are currently unknown, and their prolonged use is not recommended until more data are available.

### Long-Chain Essential Fatty Acids

Oral administration of  $\gamma$ -linoleic acid (GLA)-rich primrose oil resulted in significant improvement of UP in chronic dialysis patients. Supplementation of GLA-rich primrose oil is thought to augment synthesis of antiinflammatory eicosanoids. Similar effects could be obtained through use of fish oil, olive oil, and safflower oil.

### Capsaicin

Capsaicin (*trans*-8-methyl-*N*-vanillyl-6-nonenamide) is a natural alkaloid found in the pepper plant that depletes the cutaneous type C sensory nerve endings of substance P. Two clinical studies showed that application of 0.025% capsaicin cream significantly alleviated UP in dialysis patients, who exhibited no side effects.

### Oral Activated Charcoal

UP symptoms completely disappeared or were significantly reduced in chronic dialysis patients treated with activated charcoal (6 g/day) for 8 weeks. In two different clinical studies, comparable results were obtained with this inexpensive and well-tolerated compound, rendering it a valuable alternative for patients with UP.

### Miscellaneous

Various other therapies have been examined for the treatment of UP; some may be effective but are not first-line treatments in chronic HD patients because of undesirable side effects, cumbersome use, or incompatibility with kidney replacement therapy (sauna, cholestyramine, nicergoline). Other therapies that have not reduced UP convincingly in controlled studies include acupuncture, turmeric, low-protein diet, intravenous (IV) lidocaine, melatonin, sertraline, oral montelukast, zinc sulfate, topical vitamin D, and mexiletine.

## BULLOUS DERMATOSES

Bullous dermatoses are reported in up to 16% of patients on maintenance dialysis. This skin disease entity mainly includes pseudoporphyrias (e.g., secondary to nonporphyrinogenic drugs and chemicals) and other photodermatoses, whereas true porphyrias (e.g., porphyria cutanea tarda [PCT], variegata porphyria) remain rare.<sup>13</sup> Pseudoporphyrias, true porphyrias, and photodermatoses are clinically and histologically similar and are characterized by a blistering photosensitive skin rash. The dorsal hands and the face are the most affected areas (Fig. 91.4).

PCT is caused by abnormalities in the porphyrin-heme biosynthetic pathway, leading to an accumulation of highly carboxylated uroporphyrins in the plasma and skin. Phenotypic expression of the disease also requires one or more of a number of external contributory factors, including alcohol, estrogens, iron overload, and infection with hepatitis B and C viruses. It is important to distinguish PCT from other porphyrias in which patients are at risk for development of potentially fatal neurologic attacks if they are exposed to porphyrinogenic drugs and other precipitants. Therapeutic options include avoidance of environmental triggers, HD with high-flux membranes, repeated small-volume phlebotomies, and iron chelators.

The term *pseudoporphyria* was originally used for patients with normal plasma porphyrins who exhibited PCT-like skin lesions secondary

to drugs and chemicals. However, some dialysis patients also develop similar skin lesions that spontaneously heal and leave a hypopigmented area; this entity is known as *dialysis porphyria*; a proportion of these patients have raised plasma porphyrins but without the disturbances in porphyrin metabolism classically found in the porphyrias. In rare patients, an offending medication can be identified. However, in most dialysis patients, protection from sun exposure appears to be the only preventive measure.

## CALCIFIC UREMIC ARTERIOLOPATHY (CALCIPHYLAXIS)

### Definition

CUA, or calciphylaxis, is a devastating and life-threatening ischemic vasculopathy seen primarily among patients with CKD.<sup>14,15</sup> The ischemia may be so severe that frank infarction of downstream tissue develops. The most common and most noticeable damage is in the skin and subcutaneous tissues.<sup>12,13</sup> CUA should be distinguished from benign nodular calcification (calcinosis cutis), which can develop in patients with very high serum calcium-phosphate product (Fig. 91.5).

### Epidemiology, Pathogenesis, and Risk Factors

Although hard epidemiologic data are lacking, it is generally believed that the incidence of CUA is increasing. This might, in part, result from increased physician awareness and the high-risk profile of contemporary dialysis patients.<sup>15</sup> The estimated incidence ranges from 1 to 4 per 100 patient-years. The pathogenesis of CUA is complex and incompletely understood.<sup>14,16</sup> Recent data suggest that CUA involves a cell-mediated, bone morphogenetic protein-2–driven osteogenic process with extensive subcutaneous extracellular matrix remodeling and deposition of hydroxyapatite.<sup>17</sup> A cascade consisting of matrix remodeling, calcification, endothelial damage and thrombus formation, luminal obstruction, and finally full-blown CUA has been postulated.<sup>17</sup> Female sex, white ethnicity, obesity, diabetes mellitus, long dialysis vintage, and vitamin K deficiency are established risk factors.<sup>15</sup> In a case control study, the incidence of CUA was 10-fold higher in dialysis patients treated with vitamin K antagonists (warfarin or coumarins). Endogenous inhibitors of vascular calcification, such as matrix Gla protein, are dependent on vitamin K for their activation. Patients treated with warfarin or coumarins may thus not be able to appropriately inhibit vascular calcification. CUA prevalence is higher in patients with hyperphosphatemia and/or hypercalcemia. In acute CUA, serum calcium and phosphate levels can be low. Although data from the Evaluation of Cinacalcet HCl Therapy to



**Fig. 91.4** Porphyria Cutanea Tarda. Tense bullae, erosions, and crusts of the dorsal hands. (From Kuypers DR, Claes K, Evenepoel P, et al. A prospective proof of concept study of the efficacy of tacrolimus ointment on uraemic pruritus [UP] in patients on chronic dialysis therapy. *Nephrol Dial Transplant*. 2004;19(7):1895–1901.)



**Fig. 91.5** Benign nodular calcification (calcinosis cutis). Firm subcutaneous nodule adjacent to the elbow.





**Fig. 91.6** Proximal (A) and distal (B–C) calcific uremic arteriopathy.

Lower Cardiovascular Events (EVOLVE) trial indicate cinacalcet may reduce the incidence of CUA in HD patients who have moderate to severe secondary hyperparathyroidism,<sup>18</sup> registry data fail to confirm a key role for uncontrolled hyperparathyroidism in the pathogenesis of CUA.<sup>19</sup> Finally, both low- and high-turnover bone disease have been associated with CUA.

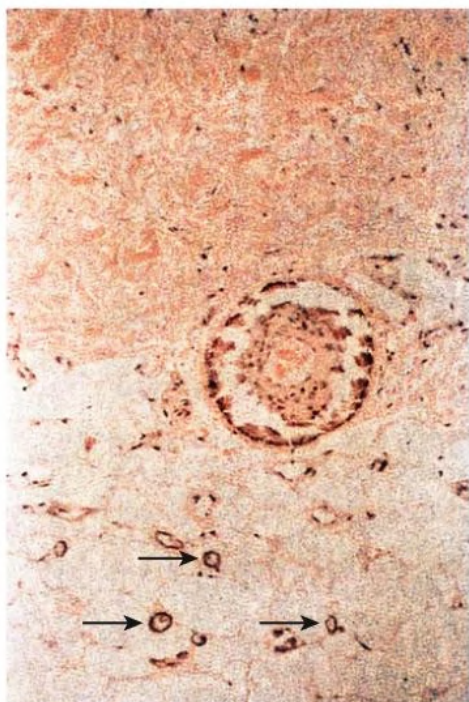
### Clinical Manifestations

CUA is typically characterized by areas of zoster-like tenderness and ischemic necrosis of the dermis, subcutaneous fat, and, less often, muscle.<sup>14</sup> These ischemic changes lead to livedo reticularis or violaceous, painful, plaque-like subcutaneous nodules on the trunk, buttocks, or proximal extremity—that is, in areas of greatest adiposity (proximal CUA; Fig. 91.6A). The early purpuric plaques and nodules progress to ischemic or necrotic ulcers with eschars that often become infected. Proximal CUA is frequently precipitated by a specific event, such as

local skin trauma or a hypotensive episode. CUA also can affect the hands, fingers, and lower extremities, thereby mimicking atherosclerotic peripheral vascular disease (distal CUA; Fig. 91.6B). Peripheral pulses are preserved distal to the area of necrosis. Myopathy, hypotension, fever, dementia, and infarction of the central nervous system, bowel, or myocardium have been described in association with cutaneous necrosis. This condition is termed *systemic CUA*.

### Pathology

Characteristic histologic features of CUA include calcification, fibrointimal hyperplasia, and thrombosis in microvessels in the subcutaneous adipose tissue and dermis, often accompanied by necrosis and adipose tissue, dermal-epidermal separation, panniculitis, proliferation of dermal epithelial cells, and interstitial deposits<sup>15</sup> (Fig. 91.7). Calcified lesions consist of hydroxyapatite, typically occupy the entire circumference of the vessel, and are mainly located in the intima.<sup>20</sup>



**Fig. 91.7 Histopathologic Features of Calcific Uremic Arteriopathy.** Medial calcification and intimal hyperplasia of an arteriole at the dermal-subcutaneous junction. Note calcification of interlobular capillaries (arrows) in the subcutaneous tissue. (Van Kossa stain.) (From Kuypers DR, Claes K, Evenepoel P, et al. A prospective proof of concept study of the efficacy of tacrolimus ointment on uraemic pruritus [UP] in patients on chronic dialysis therapy. *Nephrol Dial Transplant*. 2004;19[7]:1895–1901.)

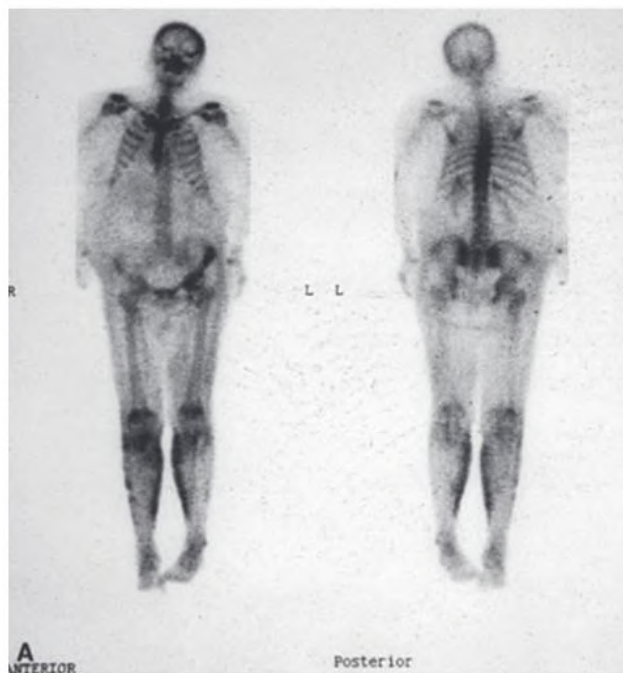
## Diagnosis and Differential Diagnosis

Many clinicians base the diagnosis of CUA on physical examination findings only. Although ulceration is an obvious presentation of CUA, increasing awareness of the condition should allow diagnosis at an earlier, nonulcerative stage when a distinct subcutaneous tenderness can be felt below the early skin lesions. Biopsies are discouraged by most but not all experts because of potential ulceration in the region of the incision and the risk for sampling error. Other potentially useful diagnostic procedures include measurements of transcutaneous oxygen saturation, bone scintigraphy (commonly with Tc-99m-methylene diphosphonate) (Fig. 91.8), and xeroradiography.<sup>21</sup>

The following conditions should be considered in the differential diagnosis: herpes zoster, systemic vasculitis, peripheral vascular disease, pyoderma gangrenosum, atheroemboli, cryoglobulinemia, and systemic oxalosis.<sup>15</sup> Skin biopsy is the standard method for confirmation of clinically suspected calciphylaxis; however, its role in practice is debated, given the risk of provoking new, nonhealing ulcers and infection. A skin biopsy should therefore be performed only when clinical circumstances do not suggest CUA or when clinical and laboratory examinations, including the assessment of coagulation and immunologic parameters, point to an alternative diagnosis.<sup>15</sup>

## Natural History

Despite intensive combined treatments, the prognosis of patients with CUA remains poor; the overall 1-year survival is 45% and the 5-year survival is 35%, with a relative risk for death of 8.5 compared with



**Fig. 91.8 Bone Scintigraphic Abnormalities in Calcific Uremic Arteriopathy.** Calf calcification in a patient with gross ulcerations in both legs from the popliteal fossae to the ankles. (From Nigwekar SU, Thadani R, and Brandenburg V. Calciphylaxis. *N Engl J Med*. 2018;378:1704–1714.)

that for other dialysis patients. Patients with ulcerative or proximal CUA have the worst prognosis. Infection accounts for up to 60% of the mortality.<sup>14</sup>

## Prevention and Treatment

Preventive approaches include controlling bone and mineral metabolism and optimizing nutritional state.<sup>14</sup> Specific therapeutic regimens have been limited to uncontrolled case series (Box 91.1). Intervention should include an aggressive program of wound care and prevention of superinfection, adequate pain control, and correction of underlying abnormalities in serum calcium and phosphorus concentrations. This includes cessation of vitamin D supplementation, intensification of the dialysis regimen, and use of a low-calcium dialysate and non-calcium-containing phosphate binders. Furthermore, local tissue trauma, including subcutaneous injections, should be avoided. Parathyroidectomy or calcimimetic agents should be reserved for patients with severe hyperparathyroidism. Vitamin K (preferentially vitamin K<sub>1</sub>) supplementation is advised, especially in patients with warfarin- or coumarin-associated CUA. Beneficial outcomes have been reported with bisphosphonates in case reports and small series. Sodium thiosulfate has been licensed as an orphan drug for CUA by the European Medicines Agency and is also approved for use in the U.S. It enhances the solubility of calcium deposits<sup>22</sup> because exchange of calcium for sodium results in extremely soluble calcium thiosulfate. Besides being a chelator of calcium, sodium thiosulfate is also a potent antioxidant. Sodium thiosulfate is given via IV at the end of every HD session (12.5–25 g over 30–60 minutes). Apart from nausea and vomiting, the therapy is well tolerated. Major side effects of sodium thiosulfate infusion include metabolic acidosis, volume overload, hypocalcemia, and QT-interval prolongation. The optimal duration of treatment and potential effects of long-term treatment on bone are unknown. Efficacy has been suggested by several case-series, but definite proof by a randomized controlled trial is lacking.



### BOX 91.1 Treatment Options in Patients With Calcific Uremic Arteriopathy

1. Reduction of procalcifying factors
  - Intensified dialysis (e.g., daily hemodialysis, switch from peritoneal dialysis to hemodialysis, low-calcium dialysate)
  - Avoidance of vitamin D and calcium supplements; administration of calcium-free phosphate binders; bisphosphonate administration (caution if adynamic bone disease is suspected)
  - Parathyroidectomy (in the case of hyperparathyroidism) or administration of cinacalcet
2. Improving the status of calcification inhibitors
  - Halt vitamin K antagonists (warfarin)
  - Aggressive treatment of infections or other proinflammatory stimuli to increase fetuin-A levels
  - Experimental:
    - Administration of high-dose vitamin K2, and, if unavailable, vitamin K1?
    - Administration of fetuin-A (e.g., by fresh frozen plasma or plasma exchange?)
3. Prevention or reversal of calcium-phosphate precipitation
  - Administration of sodium thiosulfate
4. Supportive measures
  - Avoidance of additional local tissue trauma by atraumatic wound care with gentle debridement of necrotic tissue and avoidance of subcutaneous injections
  - Anticoagulation (heparin and low molecular weight heparin)
  - Adequate pain management
  - Adequate infection control

Modified from Colboc H, Moguelet P, Bazin D, et al. Localization, morphologic features, and chemical composition of calciphylaxis-related skin deposits in patients with calcific uremic arteriopathy. *JAMA Dermatol.* 2019;155(7):789-796.

SNF472, an IV formulation of myo-inositol hexaphosphate, is currently being investigated in CUA patients (NCT04195906). SNF472 acts through a novel pathway to selectively and directly inhibit the formation and growth of hydroxyapatite crystals, the final common step in the pathophysiology of vascular calcification. SNF472 significantly attenuates the progression of coronary artery calcium and aortic valve calcification in patients with ESKD receiving HD.<sup>23</sup>

## NEPHROGENIC SYSTEMIC FIBROSIS

### Definition

NSF, formerly known as *nephrogenic fibrosing dermopathy*, is a scleroderma-like fibrosing disorder that develops in the setting of kidney failure. The fibrotic process affects the dermis, subcutaneous tissues, fascia, and other organs, including striated muscles, heart, and lungs.<sup>24</sup>

### Pathogenesis

Gadolinium-based contrast (GBC) agents have been identified as the cause of NSF; exposure to gadolinium before the onset of disease was confirmed in more than 95% of reported cases. Free gadolinium ions are highly toxic to tissues. The toxic effects of gadolinium are circumvented by sequestration of the metal by chelates, large organic molecules that form a stable complex with gadolinium and make the ion biochemically inert and nontoxic. Under normal circumstances, GBC agents are eliminated by the kidney through glomerular filtration. Evidence points toward aberrant activation of circulating fibrocytes as a central event in the genesis of NSF. Other investigators have

suspected that the strongly profibrotic mediator transforming growth factor- $\beta$  may be involved in the pathogenesis of NSF.

### Epidemiology

NSF is a rare disorder. Since the identification of the first patients with NSF in 1997, the NSF registry ([www.icnfd.org](http://www.icnfd.org)) has confirmed more than 380 patients from medical centers worldwide. NSF equally affects males and females. The risk for development of NSF after GBC agent exposure is related to the degree of kidney failure and stability of the chelate. The incidence of NSF in patients with severe kidney dysfunction (glomerular filtration rate [GFR] < 30) varies from 0.2% to 4%. Gadodiamide (marketed as OmniScan in the United States), the linear nonionic chelate-based formulation, appears to be associated with the highest risk for NSF. Gadopentetate, the linear ionic chelate-based product, probably has a medium risk, less than linear nonionic chelates but more than macrocyclic chelates. Other factors reported to be associated with NSF (without definitive proof) include coagulation abnormalities and deep venous thrombosis, recent surgery (particularly vascular surgery), hyperphosphatemia, and the use of high doses of recombinant erythropoietin. Angiotensin-converting enzyme inhibitors might protect against NSF. After warnings issued by the U.S. Food and Drug Administration (in 2007) and European Medicines Agency regarding the association of linear GBC agents with NSF in individuals with kidney dysfunction, the incidence of NSF declined and in recent years, no new cases of NSF have been reported.<sup>25</sup>

### Clinical Manifestations and Natural History

The lesions of NSF are typically symmetric and develop on limbs and trunk. A common location is between the ankles and mid thighs and between the wrists and mid-upper arms bilaterally. On occasion, swelling of the hands and feet, sometimes associated with bullae, is noted. The primary lesions are skin-colored to erythematous papules that coalesce into erythematous to brawny plaques with a peau d'orange appearance (Fig. 91.9A). These plaques have been described as having an ameboid advancing edge. Nodules are sometimes also described. Involved skin becomes markedly thickened and woody in texture. Joint contractures may develop rapidly, with patients becoming wheelchair-dependent within days to weeks of onset (see Fig. 91.9B). Patients often report pruritus, causalgia, and sharp pains in the affected areas.<sup>24</sup> Although NSF has not been reported as a cause of death, this disorder has led to reduced mobility and/or superinfection that ultimately resulted in a protracted hospital course and death.

### Pathology

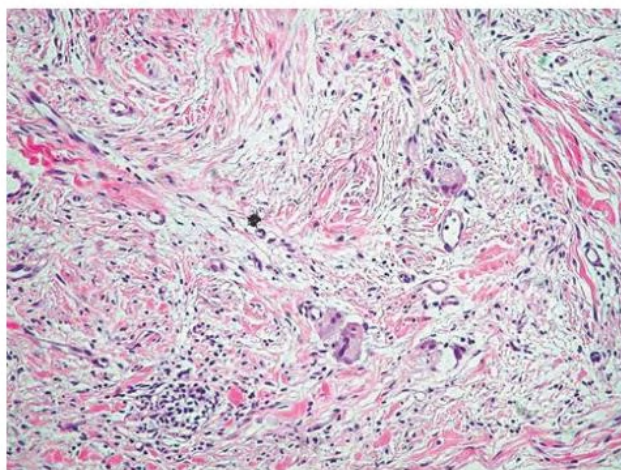
The histopathologic changes in affected skin include marked proliferation of spindle cells, the presence of numerous dendritic cells, and the accumulation of mucinous material and thick collagen bundles (Fig. 91.10). Most dermal spindle cells in NSF have the immunophenotype of a circulating fibrocyte, a recently characterized circulating cell that expresses markers of both connective tissue cells and circulating leukocytes. Metastatic calcification and NSF may be found in the same lesion.

### Diagnosis and Differential Diagnosis

The gold standard of diagnosis is histopathologic examination of skin biopsy specimens from an involved site. Skin lesions also can be visualized by [<sup>18</sup>F]-fluorodeoxyglucose whole-body positron emission tomography (PET). NSF resembles other fibrosing skin disorders, including scleromyxedema, scleroderma, eosinophilic fasciitis, eosinophilia-myalgia syndrome, and Spanish toxic oil syndrome. The specific distribution of cutaneous involvement, the occurrence in the



**Fig. 91.9** Nephrogenic Systemic Fibrosis. (A) Peau d'orange appearance. (B) Swelling of the hands, accompanied by palmar erythema, blisters, and contracture of the fingers.



**Fig. 91.10** Histopathologic Features of Nephrogenic Systemic Fibrosis. Haphazardly arranged dermal collagen bundles with surrounding clefts and a strikingly increased number of similarly arranged spindle-shaped and plump fibroblast-like cells.

setting of kidney failure, the history of recent exposure to linear GBC agents, and the unique histopathologic features distinguish NSF from the other fibrotic disorders.

### Treatment and Prevention

There is variable evidence for the efficacy of plasma exchange. Other therapeutic modalities that have been used (or are under investigation) include imatinib, oral and topical corticosteroids, selective histamine blockade, calcipotriene ointment, cyclophosphamide, cyclosporine, thalidomide, interferon- $\alpha$ , photopheresis, and psoralen ultraviolet A therapy. Intensive physiotherapy is advised in every patient to prevent or to reverse contractures of the joints. At present, prevention of NSF seems more important than any of the currently available interventions, and widespread clinical awareness of this condition is required. Avoidance of linear GBC agents in high-risk patients (acute kidney injury and patients with CKD with estimated GFR [eGFR] rate  $< 30$  mL/min/1.73 m<sup>2</sup>) is the best measure to prevent this catastrophic complication.<sup>25</sup> According to current guidelines, group II GBC agents, including three macrocyclic agents with 100% kidney excretion (gadoteridol, gadoterate meglumine, and gadobutrol) and one linear ionic agent with approximately 95% kidney and 5% hepatobiliary excretion (gadobenate dimeglumine) can be administered safely independent of preprocedural kidney function screening. Dialysis initiation or alteration is likely unnecessary based on group II or group III GBC administration.<sup>26</sup>

## SELF-ASSESSMENT QUESTIONS

- UP in dialysis patients is:
  - a local skin disorder.
  - a systemic disorder caused by uremia.
  - a neurologic disorder caused by overexpression of opioid skin receptors.
  - an inflammatory systemic disorder caused by dysregulation of the immune system.
  - all of the above.
- Skin emollients in dialysis patients should be used:
  - only in patients without underlying primary skin diseases.
  - only in patients with proven xerosis.
  - only in patients without skin defects.
  - only in patients with therapy failure.
  - in all patients.
- Treatment of UP with oral nalfurafine:
  - is limited to patients with proven overexpression of  $\mu$  receptors in their skin.
  - is limited to patients free from sleeping disturbances.
  - is limited to patients with therapy failure.
  - is limited to patients with baseline normal blood prolactin concentrations.
  - is not limited to any subgroup of patients with UP.



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4. Which of the following drugs predisposes to calcific uremic arteriopathy?
- A. Acetylsalicylic acid
  - B. Heparin
  - C.  $\beta$ -Blocker
  - D. Warfarin and coumarins
  - E. Calcium-free phosphate binders
5. Nephrogenic systemic fibrosis:
- A. is a self-limiting disease with benign prognosis.
  - B. is related to exposure to linear gadolinium chelates.
  - C. is painless.
  - D. is especially prevalent in kidney transplant recipients.
  - E. is related to exposure to iodine radiocontrast agents.
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